

01_MasterSpec.md

MonitorAI v1.5 Master Specification

Version: 1.5

Status: Specification Baseline

Document Type: Master Specification

1. Purpose

MonitorAI is an orthodontic image acquisition platform designed to collect highly standardized intraoral photographs for:

- Clinical review
- Remote monitoring
- Future AI dataset generation

MonitorAI is not intended to diagnose patients.

MonitorAI is intended to acquire reproducible clinical records.

2. Mission

Create the highest quality orthodontic image dataset possible while maintaining a workflow that patients can complete independently.

The system must balance:

- Image Quality
- Metadata Quality
- Patient Compliance

The system should never optimize one of these at the expense of the others.

3. Clinical Philosophy

Technology exists to improve diagnosis.

Diagnosis exists to improve treatment.

Treatment exists to improve patient outcomes.

MonitorAI therefore prioritizes:

1. Standardization
2. Repeatability
3. Clinical Usefulness
4. Metadata Quality
5. Future AI Value

over implementation convenience or visual complexity.

4. Product Definition

MonitorAI is not:

- A photography application
- A telemedicine platform
- A diagnostic AI

MonitorAI is:

A standardized orthodontic image acquisition platform.

5. Dataset Philosophy

The primary long-term asset of MonitorAI is not software.

The primary long-term asset is the dataset.

Every image should be considered a future training sample.

Every design decision should be evaluated by asking:

"Will this improve future dataset quality?"

6. Metadata Philosophy

Metadata is equal in importance to image data.

Every image must have associated metadata.

Image without metadata is considered incomplete.

Metadata must be captured automatically whenever possible.

7. Clinical Visibility Philosophy

MonitorAI prioritizes visibility over geometric positioning.

The goal is not:

Perfect camera angle.

The goal is:

Clinically useful anatomy visibility.

Posterior visibility is prioritized.

Occlusal visibility is prioritized.

Clinical usefulness is prioritized.

8. PTG Philosophy

PTG (Progressive Target Guidance) exists to guide patients toward clinically useful images.

PTG evaluates:

- Posterior tooth visibility
- Occlusal arch visibility
- Clinical usefulness

PTG should not depend primarily on head rotation measurements.

9. Capture Philosophy

Patients should not need photographic skills.

The application is responsible for:

- Guidance
- Validation
- Capture timing

Patients are responsible only for:

- Following instructions
- Positioning themselves

Automatic capture is preferred over manual capture.

10. Retake Philosophy

Retakes should be minimized.

However:

Poor quality images should never be accepted solely to reduce retakes.

The system should seek the best balance between:

- Completion rate
 - Image quality
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11. Device Philosophy

The platform is designed for iPhone.

Minimum Supported Device:

iPhone 11

Primary Development Device:

iPhone 15 Pro Max

Supported Scope:

All compatible iPhone 11–17 generation devices.

Device-specific workflows should be avoided.

12. Workflow Philosophy

The workflow must be:

- Predictable
- Repeatable
- Simple

Every patient should follow the same sequence.

Standardization is preferred over customization.

13. AI Strategy

MonitorAI is a dataset generation platform.

Future AI applications may include:

- Aligner fit assessment
- Tooth movement analysis
- Monitoring automation
- Clinical alerts
- Decision support

The current version does not perform these tasks.

The current version collects the data required for these future systems.

14. Security Philosophy

Patient privacy is mandatory.

All communication must be encrypted.

Sensitive information must never be exposed through:

- Logs
- Analytics
- Debug output

Security is a core requirement.

15. Reliability Philosophy

Images must not be lost.

Metadata must not be lost.

Sessions must not be lost.

Offline operation must be supported whenever possible.

Reliability is more important than convenience.

16. Development Philosophy

Code should be:

- Testable
- Modular
- Maintainable
- Documented

Temporary fixes are prohibited.

Root-cause resolution is required.

17. Definition of Success

MonitorAI is successful when:

Patients can independently create clinically useful images.

Clinicians can reliably review those images.

The resulting dataset can support future AI development.

18. Final Principle

MonitorAI is not a camera application.

MonitorAI is a standardized orthodontic image acquisition platform and future AI dataset generation platform.

Every design decision should prioritize:

- Standardization
- Repeatability
- Clinical usefulness
- Metadata quality
- Future AI value